

# SAATHI — Final Integration

Complete integration plan for the four independent AI pipelines

## Purpose

• Integrate the four independently developed pipelines into one reproducible SAATHI intelligence stack. Integration happens only after each team passes its acceptance tests.

## Step 1 — Freeze interfaces

- Freeze the canonical corpus schema from Team 1.
- Freeze Team 2's `retrieve_evidence(query, filters)` contract.
- Freeze Team 3's `generate_grounded_answer(query, evidence, language)` contract.
- Freeze Team 4's evaluation and intelligence data contracts.

## Step 2 — Assemble the knowledge layer

- Load Team 1's corpus and version registry.
- Build/load Team 2's vector and BM25 indexes against exactly that corpus version.
- Verify every vector maps to a valid chunk and every chunk maps to a source.

## Step 3 — Assemble the RAG path

- User question → query processing → hybrid retrieval → reranking → evidence selection → RAGA model → grounding validator → citation mapper → response.
- Log every stage with a request ID so failures can be diagnosed.

## Step 4 — Assemble advanced intelligence

- Connect classification, scheme selection, compliance chain, lab matching, regulatory change intelligence and Cortex to the same evidence layer.
- Do not create duplicate corpora or separate citation systems.

## Step 5 — End-to-end evaluation

- Run the frozen golden dataset.
- Compare the integrated system with the original baseline.
- Report retrieval metrics, answer metrics, citation correctness, unsupported-answer rate and latency.

## Step 6 — Failure triage

- If evidence is wrong, send the issue to Team 1/2.
- If evidence is correct but answer is wrong, send it to Team 3.
- If the evidence and answer are correct but the workflow is wrong, send it to Team 4.
- Do not fix every failure with prompt changes.

## Step 7 — Demo hardening

- Freeze the demo corpus and model versions.
- Pre-warm model/index loading.

- Test network/API failures and model timeouts.
- Keep a controlled fallback path that never invents regulatory facts.
- Prepare 5–10 verified demo scenarios with expected outputs and source evidence.

## Step 8 — Final architecture

- USER → Query Processing → Hybrid Retrieval (Vector + BM25) → Reranker → BIS Evidence → RAGA/LLM → Grounding Validator → Answer + Citations.
- Advanced modules consume the same evidence and provenance layer.
- Evaluation continuously measures retrieval and answer quality.

## Final deliverables

- one reproducible repository
- one frozen corpus version
- one model/version registry
- one retrieval stack
- one RAG pipeline
- one citation/grounding system
- one evaluation suite
- deployment instructions
- demo runbook

## Final Definition of Done

- Controlled 600+ MB government-source corpus is indexed with provenance.
- Current/superseded versions are represented.
- Hybrid retrieval and reranking work.
- The 1.25 GB model is benchmarked and used only where its actual role justifies it.
- Answers are evidence-backed and citations resolve to real chunks.
- Unsupported questions decline safely.
- English and Hindi flows pass evaluation.
- Cortex and C5–C8 consume the same evidence layer.
- Metrics are reproducible and demo telemetry is not mislabeled as production accuracy.